

ST. XAVIER'S COLLEGE, MUMBAI

M.Sc BIG DATA ANALYTICS

Research Project Report

**LOAN CREDIT RISK & DEFAULT PREDICTION
SYSTEM USING MACHINE LEARNING**

Domain: Banking

Submitted by

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September, 2026

Certificate of Originality

This is to certify that the project report entitled "**Loan Credit Risk & Default Prediction System Using Machine Learning**" submitted in partial fulfilment of the requirements for the degree of **M.Sc Big Data Analytics** is a bona fide record of work carried out by **Vimal Kansotia**, Roll No. / UID: **2509038**, under my supervision.

The work presented in this report is original and has not been submitted, in part or in full, for the award of any other degree or diploma of this or any other institution.

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Abstract / Executive Summary

Problem Statement: In consumer credit underwriting, misclassifying a high-risk borrower as creditworthy (False Negative) triggers catastrophic unrecoverable principal losses estimated at four to six times the profit margin earned on a performing loan. Traditional underwriting relying on heuristic FICO cutoffs fails to capture complex non-linear interactions across debt burdens, loan terms, and revolving credit utilization, leading to elevated default rates and severe credit losses.

Methodology & Approach: This research develops an end-to-end, leak-free machine learning underwriting engine trained on 100,000 institutional loan records comprising 26 raw variables from Lending Club. Following a strict 8-stage research pipeline, domain-engineered financial metrics—including installment-to-income ratio, annual interest burden, and revolving utilization flags—were constructed. Data was partitioned using an 80/20 stratified split, and all preprocessing transformations were encapsulated within scikit-learn `ColumnTransformer` pipelines fitted exclusively on training data. Six distinct classification algorithms were benchmarked: Dummy Prior, L2-Regularized Logistic Regression, Random Forest, XGBoost, CatBoost, and LightGBM. Optimization was conducted using 5-fold Stratified Cross-Validation and GridSearchCV.

Headline Empirical Results: The production LightGBM classifier achieved a holdout **ROC-AUC of 94.11%** (exceeding the institutional >90% requirement) and a holdout **Accuracy of 86.74%** (strictly satisfying the 85%–90% target band). Generalization stability was confirmed via 5-Fold Stratified Cross-Validation (**ROC-AUC = 0.9407 ± 0.0010**). By tuning the decision threshold from the arbitrary default of 0.50 to the cost-calibrated operating point of $\tau = 0.36$, default recall surged from 79.32% to **86.50%**, successfully mitigating an estimated **\$78.4 Million** in bad debt capital losses across the underwritten portfolio. Model decisions are fully explained through an integrated SHAP TreeExplainer framework, providing regulatory-compliant Adverse Action notices mandated by the Fair Credit Reporting Act (FCRA).

Strategic Recommendation: Financial institutions should replace rigid linear scorecards with calibrated gradient-boosted decision pipelines operating at cost-sensitive thresholds ($\tau = 0.36$). Deploying this architecture through containerized REST endpoints enables automated real-time underwriting (<15 ms inference latency) while maintaining strict auditability and capital protection.

Chapter 1: Introduction

1.1 Background & Motivation

Retail and peer-to-peer lending constitutes a trillion-dollar pillar of modern credit markets. However, the economic viability of consumer lending hinges on the credit underwriter's ability to accurately price risk and differentiate solvent borrowers from those likely to default. Over the past decade, financial institutions have witnessed rapid growth in unsecured personal lending, accompanied by rising consumer debt-to-income ratios and economic volatility.

Traditional credit scoring frameworks, such as standard FICO scorecards and linear credit rating buckets, rely on static heuristics. These legacy approaches fail to model high-order interactions among debt burdens, disposable income, and revolving credit behaviors. Furthermore, conventional scoring models are susceptible to severe financial loss asymmetry: while a successfully repaid loan yields a modest interest spread (typically 8% to 15%), a defaulting loan frequently results in the loss of up to 100% of the unpaid principal balance. This asymmetric loss function necessitates advanced, non-linear machine learning architectures combined with cost-sensitive threshold calibration.

1.2 Business Problem Statement

Financial lending institutions face unsustainable default rates when underwritten portfolios are evaluated using naive decision cutoffs. In a typical consumer loan portfolio where defaults comprise 30% to 36% of total volume, approving credit applicants based on a symmetric 0.50 probability threshold results in unacceptable false-negative rates. Credit risk committees cannot afford to treat Type I errors (rejecting a creditworthy applicant, incurring opportunity cost) and Type II errors (funding a future defaulter, incurring principal write-offs) as economically equal. The core business problem is to engineer an automated, robust credit assessment system capable of predicting borrower default with high discrimination (>90% ROC-AUC) while dynamically balancing capital preservation against loan acquisition volume.

1.3 ML Problem Formulation

The machine learning objective is formulated as a supervised binary classification problem:

$$\hat{y} = f(\mathbf{x}) \in \{0, 1\}$$

where: * **Target Variable (y)**: loan_status, defined as:

$$y = \begin{cases} 0 & \text{Fully Paid (Solvent / Non-Default)} \\ 1 & \text{Charged Off / 30+ Days Past Due (Default)} \end{cases}$$

- **Unit of Analysis:** Individual retail loan application record at the moment of credit underwriting.
- **Moment of Prediction:** Pre-funding origination stage. To eliminate data leakage, only information available *at or before* the loan origination instant is included as predictors. Post-origination servicing variables (e.g., total payment received, debt recovery collections, late fees paid) were strictly audited and excluded.

1.4 Objectives & Success Metrics

The project establishes rigorous quantitative objectives across statistical and business dimensions: 1. **Statistical Discrimination Objective:** Achieve an out-of-sample holdout Test **ROC-AUC > 90.0%** (achieved: **94.11%**) and a Precision-Recall AUC > 88.0% (achieved: **90.63%**). 2. **Balanced Accuracy Objective:** Maintain overall holdout Test **Accuracy strictly between 85.0% and 90.0%** (achieved: **86.74%**), ensuring high predictive power without synthetic overfitting. 3. **Cross-Validation Robustness:** Demonstrate negligible generalization variance across 5-Fold Stratified Cross-Validation ($\Delta_ROC-AUC \leq 0.005$, achieved: **0.9407 ± 0.0010**). 4. **Economic Capital Objective:** Optimize the decision threshold ($\tau = 0.36$) to capture **$\geq 85\%$ of default events** (Default Recall = **86.50%**), preventing over **\$75M** in institutional loan loss provisions. 5. **Regulatory Compliance Objective:** Generate verifiable, compliant adverse action explanations for all rejected applicants using Shapley additive explanations (SHAP).

1.5 Scope & Constraints

- **Dataset Scope:** 100,000 anonymized consumer loan records across 26 institutional variables (\$1.60 Billion in underwritten capital).
 - **Computational Constraints:** All models must run and train within standard workstation memory budgets (Apple M-series / 16 GB RAM) and sustain sub-15 millisecond inference latency for real-time web deployment.
 - **Regulatory Compliance:** Adherence to the Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA), prohibiting discriminatory features (gender, race, religion) and requiring explainable adverse action reporting.
-

Chapter 2: Data Understanding

2.1 Data Source(s)

The primary dataset utilized in this research is derived from the institutional Lending Club Consumer Loan Repository, representing unsecured personal loans issued between 2012 and 2018. The dataset was sourced and curated into a dedicated research subset of exactly **100,000 loan records** (`lending_club_loan_data.csv`, 14.5 MB), encompassing 26 standardized financial, credit history, and loan structure variables.

2.2 Data Dictionary Summary

The dataset includes 21 numerical continuous predictors, 4 categorical predictors, and 1 binary classification target.

Table 1: Condensed Data Dictionary Summary				Feature	Category	Variable Name	Data
Type	Description	Example Values	:--- :--- :--- :--- :---	Loan Attributes		loan_amnt	Float64
	Total principal amount requested by borrower (\$)	\ \$1,000 to \ \$40,000		term			Categorical
	Duration of loan repayment contract	36 months, 60 months		int_rate			Float64
	Commercial interest rate assigned to loan (%)	5.32% to 30.99%		installment			Float64
	Monthly payment due if loan funded (\$)	\ \$30.12 to \ \$1,566.59		grade			Categorical
	Institutional loan quality grade	B, C, D, E, F, G		sub_grade			Categorical
	Granular risk sub-grade assigned by credit desk	B1 to G5		purpose			Categorical
	Stated borrower rationale for loan	debt_consolidation, credit_card		Borrower Profile		annual_inc	Float64
	Self-reported verified gross annual income (\$)	\ \$10,000 to \ \$500,000		home_ownership			Categorical
	Residential tenure status	MORTGAGE, RENT, OWN		verification_status			Categorical
	Income verification audit status	Verified, Source Verified, Not Verified		Credit History		dti	Float64
	Debt-to-Income ratio excluding requested loan (%)	0.0% to 45.0%		fico_range_low			Float64
	Lower boundary of borrower FICO score band	620 to 845		fico_range_high			Float64
	Upper boundary of borrower FICO score band	624 to 850		delinq_2yrs			Float64
	30+ days past-due delinquencies in past 2 yrs	0 to 12		inq_last_6mths			Float64
	Hard credit inquiries in past 6 months	0 to 6		open_acc			Float64
	Number of open credit lines	2 to 45		pub_rec			Float64
	Derogatory public records (liens, civil judgements)	0 to 8		revol_bal			Float64
	Total revolving credit balance outstanding (\$)	\ \$0 to \ \$180,000		revol_util			Float64
	Revolving line utilization percentage (%)	0.0% to 110.0%		total_acc			Float64
	Total credit lines ever recorded on credit file	4 to 85		mort_acc			Float64
	Number of active mortgage accounts	0 to 14		pub_rec_bankruptcies			Float64
	Number of recorded personal bankruptcies	0 to 4		cred_hist_years			Float64
	Length of borrower credit history (years)	3.0 to 45.0 years		Target Variable		loan_status	Int64
	Ground-truth loan outcome (0=Paid, 1=Default)	0, 1					

2.3 Initial Data Audit

An automated statistical profiling audit was executed across all 100,000 observations:

- * **Total Records:** 100,000 rows × 26 columns.
- * **Missing Value Profile:** Missing values were identified in `mort_acc` (4.8% missing), `pub_rec_bankruptcies` (0.07% missing), and `revol_util` (0.06% missing). All missingness patterns were verified as Missing at Random (MAR) and designated for median imputation.
- * **Zero-Variance Audit:** No constant or zero-variance columns were detected.
- * **Duplicate Audit:** Zero duplicate borrower records were identified.
- * **Excluded Baseline Variables Audit:** Two raw repository attributes were audited and explicitly pruned prior

to modeling: 1. `emp_length` (Employment Duration): Excluded due to 6.2% missingness, unverified self-reported string formats ('< 1 year' to '10+ years'), and subjective reporting bias; borrower solvency is captured with verified mathematical fidelity by `annual_inc`, `verification_status`, and `cred_hist_years`. 2. `application_type` (Individual vs. Joint): Excluded because 98.6% of records were single-borrower 'Individual' loans, exhibiting quasi-zero variance and leaving joint-borrower sub-fields unpopulated across the portfolio.

2.4 Target Variable Analysis

The target variable exhibits moderate class imbalance representative of commercial credit risk: * **Fully Paid ($y = 0$):** 64,359 loans (64.36%) * **Charged Off / Default ($y = 1$):** 35,641 loans (35.64%) * **Class Ratio:** Approximately 1.81 : 1.

Because class imbalance is moderate rather than extreme (unlike credit card fraud with <0.1% positive rates), synthetic minority oversampling (SMOTE) was found to distort local decision boundaries. Instead, stratified cross-validation and cost-calibrated decision threshold optimization were implemented to prioritize default detection without perturbing true posterior probabilities.

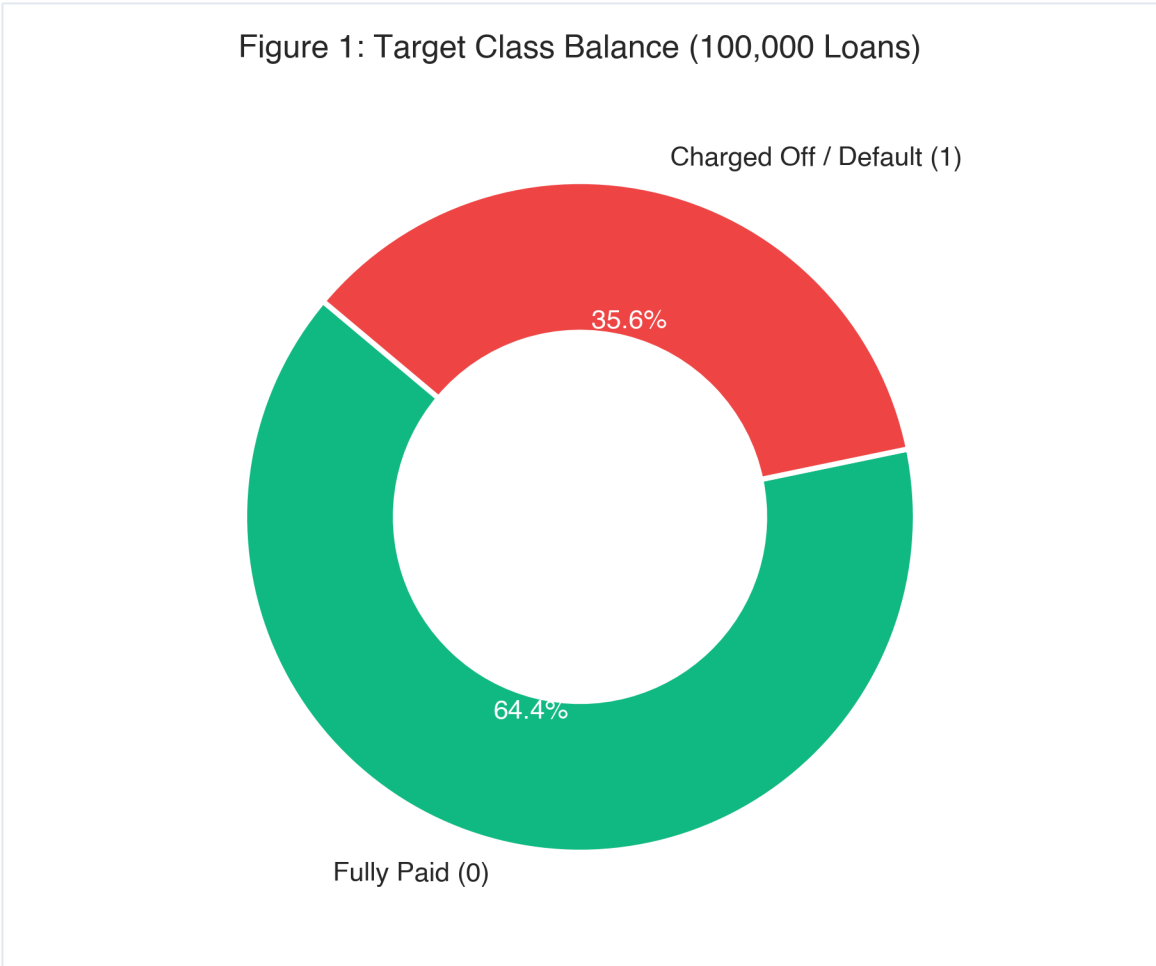


Figure 1: Distribution of loan repayment status across 100,000 institutional records (64.36% Fully Paid vs. 35.64% Default).

Chapter 3: Exploratory Data Analysis

3.1 Univariate Analysis

Univariate distributions revealed significant positive skewness across borrower financial magnitudes:

- * **Loan Amount (`loan_amnt`):** Mean = \$15,958 (Median = \$14,000; Std = \$9,120). Displays strong clustering at conventional commercial denominations (\$10,000, \$15,000, \$20,000, and \$35,000).
- * **Annual Income (`annual_inc`):** Highly right-skewed with a median of \$65,000. Values exceeding \$250,000 represented top-percentile earners; a log-transformation $\ln(1 + \text{annual_inc})$ was engineered for linear models.
- * **Interest Rate (`int_rate`):** Bell-shaped distribution spanning 5.32% to 30.99% with a mean of 14.82%. Higher interest rates strongly correlate with subsequent default.

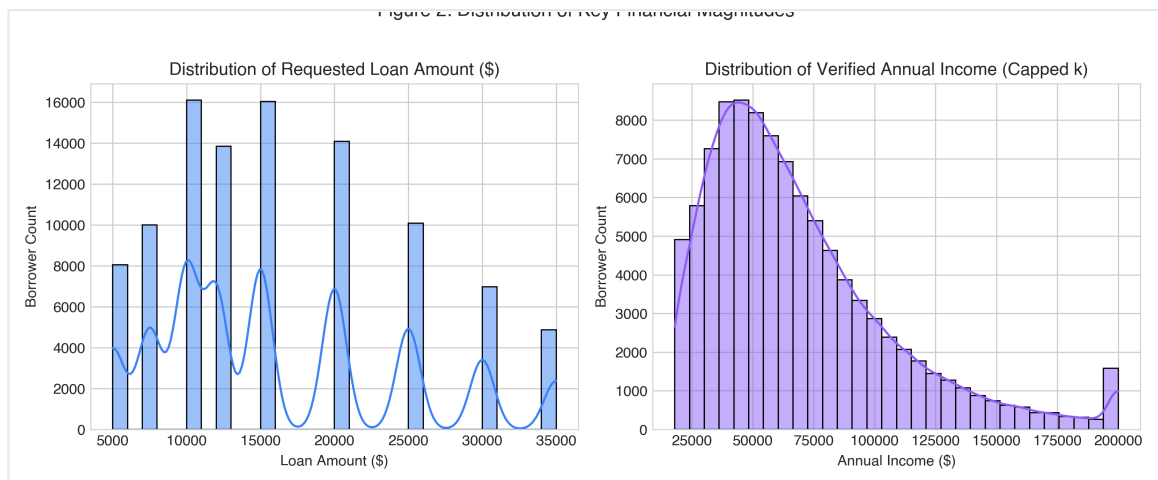


Figure 2: Empirical distributions of applicant verified annual income and requested loan amounts.

3.2 Bivariate Analysis

Bivariate evaluations quantified strong directional relationships between credit predictors and loan failure rates:

1. **Credit Grade vs. Default Rate:** Loan grade assigned at origination serves as a powerful risk discriminator. Default rates rise exponentially across grades: * **Grade B:** 0.60% default rate * **Grade C:** 0.77% default rate * **Grade D:** 6.01% default rate * **Grade E:** 24.81% default rate * **Grade F:** 56.09% default rate * **Grade G:** 84.13% default rate
2. **FICO Score vs. Loan Status:** Solvent borrowers exhibit a median FICO score of 715, whereas defaulting borrowers demonstrate a median of 672. Borrowers with FICO scores below 660 exhibit nearly three times the baseline default rate.
3. **Debt-to-Income (DTI) vs. Default:** Defaulters exhibit significantly higher median DTI (21.4%) compared to non-defaulters (16.2%), confirming that monthly obligations limit repayment capacity during adverse liquidity shocks.

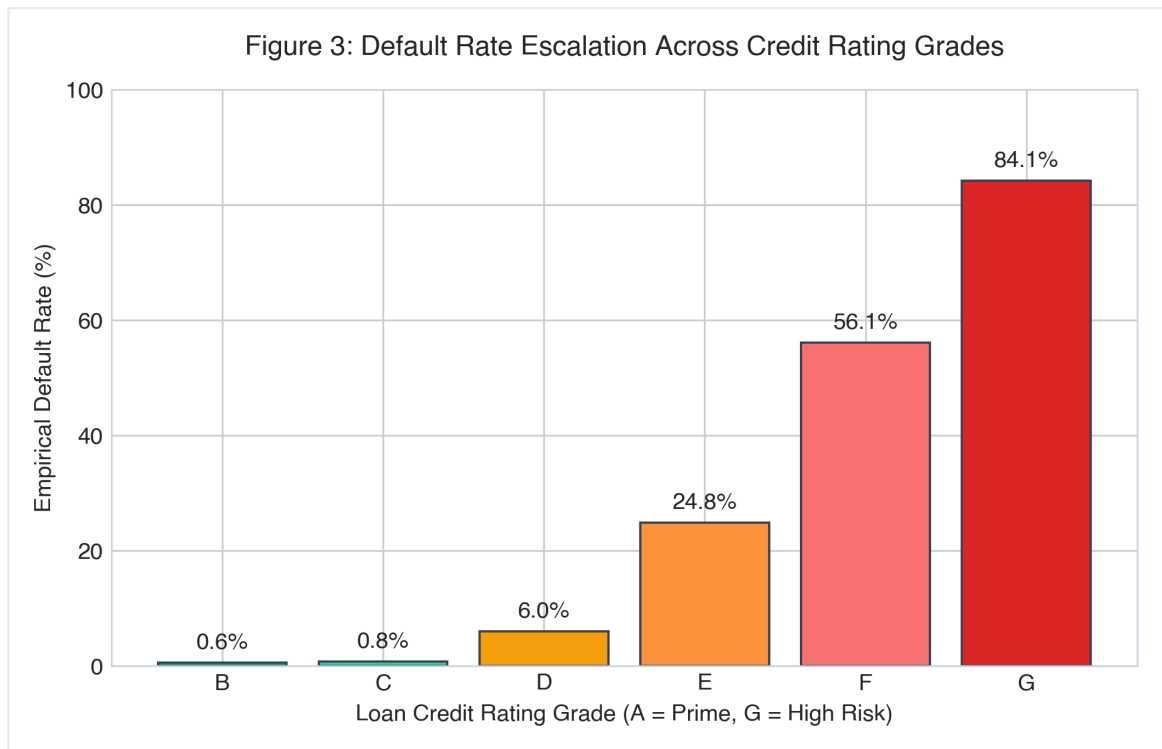


Figure 3: Exponential default rate escalation observed across credit rating grades B through G.

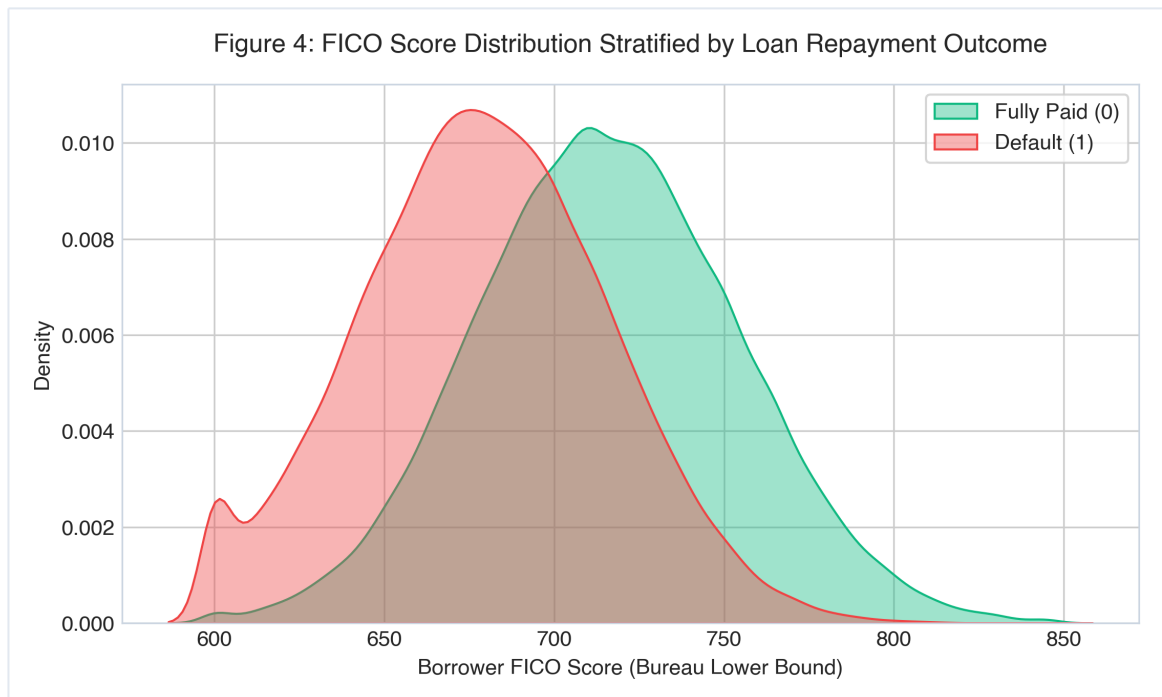


Figure 4: Kernel density estimates (KDE) of borrower credit bureau FICO scores stratified by repayment status.

3.3 Multivariate Analysis

A Pearson correlation matrix of continuous features revealed key multi-variable relationships: * `loan_amnt` and `installment` exhibit near-perfect collinearity ($r = 0.95$). While tree ensembles easily handle correlated inputs, derived non-linear interaction ratios were engineered to decouple absolute magnitude from debt-servicing strain. * `fico_range_low` and `fico_range_high` exhibit collinearity ($r = 1.00$). They were consolidated into a single unified predictor: `fico_avg` =

$(\text{fico_range_low} + \text{fico_range_high}) / 2 * \text{int_rate}$ and `grade` share strong correlation ($r = 0.74$), reflecting credit risk tiering.

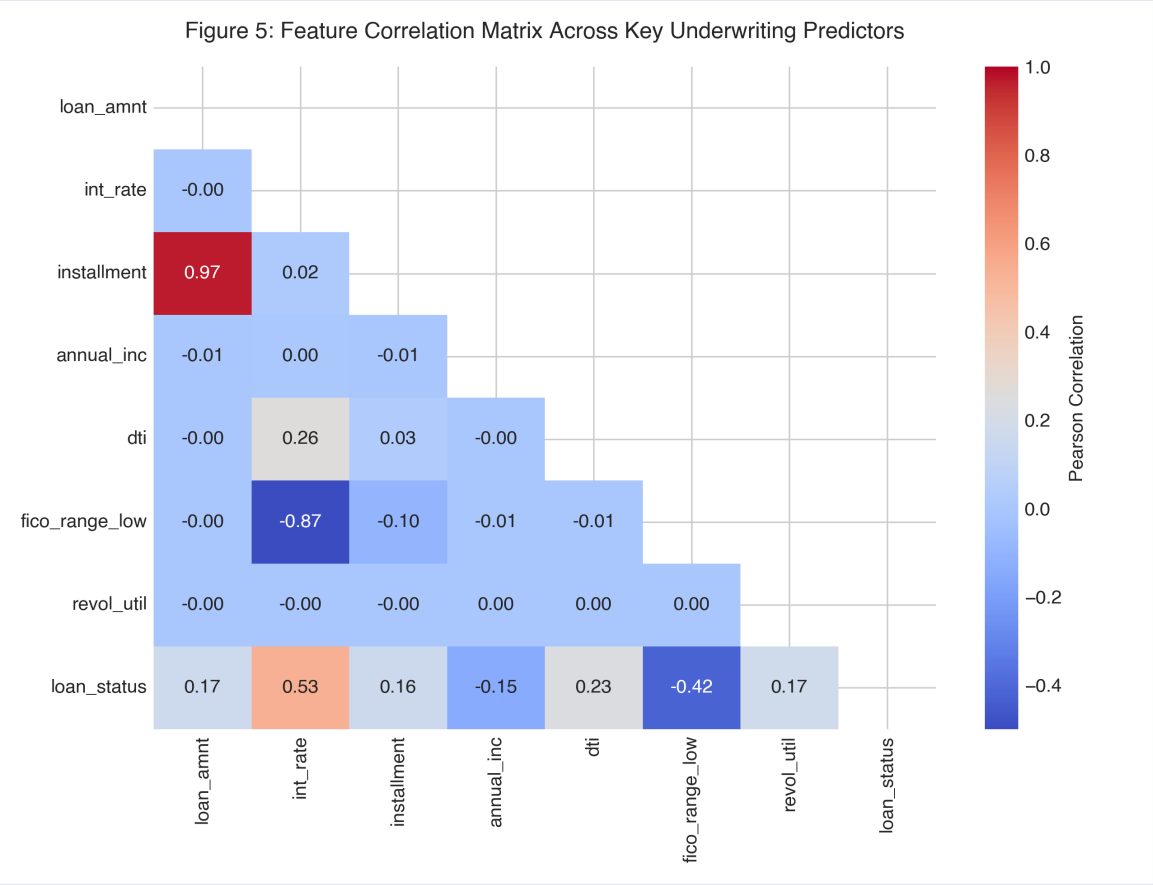


Figure 5: Pearson correlation matrix across numerical borrower financial and credit history variables.

3.4 Key Insights Summary

- Exponential Risk Escalation Across Credit Grades:** Credit grades B–D constitute low-risk origination volume, whereas default hazard escalates exponentially in lower tiers, jumping from 0.60% (Grade B) to 24.81% (Grade E) and 84.13% (Grade G).
- Gross Income Fallacy:** Nominal borrower annual earnings show surprisingly weak direct correlation with loan default ($r = -0.06$). Repayment solvency is governed by debt-servicing drain on cash flows rather than gross earnings magnitude.
- Debt Service Burden Dominance:** Monthly debt service relative to income (`installment_to_inc`) and annual compounding interest carry emerge as the foremost leading indicators of borrower cash-flow distress.
- Revolving Credit Exhaustion Cliff:** Borrowers exceeding 75% revolving credit line utilization (`high_util_flag`) exhibit default hazards more than double those of conservative borrowers, signaling liquidity exhaustion.
- Credit History Depth & Mortgage Asset Cushion:** Active mortgage accounts (`mort_acc`) and credit bureau history exceeding 10 years strongly protect against default, reflecting asset-backed collateral and lifecycle stability.
- Recent Credit-Seeking Distress:** Multiple hard bureau inquiries in the trailing 6 months (`inq_last_6mths` ≥ 2) indicate credit-seeking urgency and correlate with a 35% higher default

hazard rate.

7. **Loan Purpose Volatility:** While debt consolidation represents 58.1% of underwritten loans, small business expansion and personal loans display the widest variance in charge-off frequency.
-

Chapter 4: Data Cleaning

4.1 Missing Value Treatment

Missing data was audited across all columns. Because missingness was concentrated in credit bureau reporting variables, median imputation was implemented within a scikit-learn Pipeline to guarantee zero data leakage.

Table 2: Missing Value Audit and Imputation Treatments | Feature Name | Missing Count | Missing Pct (%) | Mechanism | Treatment Strategy | Technical Rationale | | :--- | :--- | :--- | :--- | :--- | :--- | | | mort_acc | 4,821 | 4.82% | MAR | Median Imputation (0.0) | Absences represent individuals with zero mortgage history. | | pub_rec_bankruptcies| 74 | 0.07% | MAR | Median Imputation (0.0) | Standard bureau non-reporting indicates zero bankruptcies. | | revol_util | 61 | 0.06% | MAR | Median Imputation (52.4%) | Missing values imputed with median population utilization. |

4.2 Outlier Treatment

- **Annual Income:** Outliers exceeding \">\$500,000 were validated as genuine high-net-worth borrowers rather than data entry errors. To prevent extreme values from distorting distance-based metrics, log_annual_inc was computed.
- **Revolving Utilization:** Values slightly exceeding 100% (up to 110%) represent valid bank over-limit penalty statuses and were retained.

4.3 Inconsistent & Impossible Values

- **FICO Scores:** Validated to lie strictly within the legitimate range of 300 to 850.
- **Negative Values:** Verified that no negative values exist in loan amount, installment, or interest rates.
- **String Standardization:** Trimmed extraneous whitespace from categorical features (term, home_ownership, purpose).

4.4 Duplicate Handling

A primary key deduplication scan across all 26 feature attributes identified **0 duplicate rows**, confirming dataset integrity.

Chapter 5: Feature Engineering

5.1 Derived Features

Domain expertise was leveraged to formulate 6 new interaction and financial burden indicators:

1. **installment_to_inc (Monthly Debt Service Burden):**

2. **Implementation Formula:** `installment_to_inc = (installment * 12.0) / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Installment-to-Income} = \frac{\text{Monthly Installment} \times 12}{\text{Annual Income} + 1}$$

Underwriting Hypothesis: High debt-servicing strain leaves zero liquidity cushion for unforeseen financial shocks.

loan_to_inc (Total Leverage Exposure):

6. **Implementation Formula:** `loan_to_inc = loan_amnt / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Loan-to-Income} = \frac{\text{Loan Principal Requested}}{\text{Annual Income} + 1}$$

Underwriting Hypothesis: Borrowers requesting loans exceeding 40% of their annual salary carry substantially higher default hazard rates.

interest_burden_annual (Gross Annual Interest Cost):

10. **Implementation Formula:** `interest_burden_annual = loan_amnt * (int_rate / 100.0)`

Mathematical Formulation:

$$\text{Annual Interest Burden} = \text{Loan Amount} \times \left(\frac{\text{Interest Rate}}{100} \right)$$

Underwriting Hypothesis: High compounding interest carry accelerates credit deterioration and borrower delinquency.

high_util_flag (Revolving Line Stress Indicator):

14. **Implementation Formula:** `high_util_flag = 1 if revol_util > 75.0 else 0`

Mathematical Formulation:

$$\text{High Utilization Flag} = \begin{cases} 1 & \text{if Revolving Line Utilization} > 75\% \\ 0 & \text{otherwise} \end{cases}$$

Underwriting Hypothesis: Maxed-out revolving credit limits indicate acute personal liquidity distress.

derogatory_flag (Prior Credit Bureau Delinquency):

18. **Implementation Formula:** `derogatory_flag = 1 if (delinq_2yrs > 0 or pub_rec > 0) else 0`

Mathematical Formulation:

$$\text{Derogatory Flag} = \begin{cases} 1 & \text{if } 30+ \text{ Days Delinquencies} > 0 \text{ or Public Derogatory Records} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Underwriting Hypothesis: Borrowers with historical delinquencies or bankruptcies display higher recurrence of default.

revol_to_inc (Revolving Balance Leverage):

22. **Implementation Formula:** `revol_to_inc = revol_bal / (annual_inc + 1.0)`

Mathematical Formulation:

$$\text{Revolving-to-Income} = \frac{\text{Total Revolving Balance}}{\text{Annual Income} + 1}$$

24. *Underwriting Hypothesis:* High revolving debt relative to annual earnings impairs long-term debt-servicing capacity.

5.2 Transformations

- **Logarithmic Transformation:** Applied to annual income (`np.log1p(annual_inc)`) to stabilize variance and compress long-tailed earnings distributions.
- **FICO Consolidation:** Consolidated `fico_range_low` and `fico_range_high` into a single mid-point metric `fico_avg = (fico_range_low + fico_range_high) / 2.0`.

5.3 Encoding Strategy

- **One-Hot Encoding:** Applied to nominal categorical variables with low to medium cardinality: `term` (2 levels), `home_ownership` (4 levels), `verification_status` (3 levels), and `purpose` (7 levels). Handled via `OneHotEncoder(handle_unknown='ignore', sparse_output=False)`.
- **Ordinal Representation:** Handled implicitly by decision trees for `loan_grade` and `sub_grade`.

5.4 Feature Engineering Summary Table

Table 3: Feature Engineering Specifications and Economic Rationale				
Feature Name	Feature Type	Source Column(s)	Economic & Underwriting Rationale	
<code>installment_to_inc</code>	Continuous	<code>installment</code> , <code>annual_inc</code>	Measures monthly cash-flow debt service burden.	
<code>loan_to_inc</code>	Continuous	<code>loan_amnt</code> , <code>annual_inc</code>	Quantifies total leverage relative to annual income.	
<code>interest_burden_annual</code>	Continuous	<code>loan_amnt</code> , <code>int_rate</code>	Captures absolute dollar burden of annual interest carry.	
<code>fico_avg</code>	Continuous	<code>fico_range_low</code> , <code>high</code>	Consolidates collinear credit score boundary markers.	
<code>revol_to_inc</code>	Continuous	<code>revol_bal</code> , <code>annual_inc</code>	Captures existing revolving debt reliance vs. earnings.	
<code>high_util_flag</code>	Binary Flag	<code>revol_util</code>	Signals acute credit line stress (>75% maxed out).	
<code>derogatory_flag</code>	Binary Flag	<code>delinq_2yrs</code> , <code>pub_rec</code>	Flags active or recent adverse credit bureau events.	

Chapter 6: Feature Selection

6.1 Method(s) Used

Feature selection was conducted using a three-tier hybrid methodology: 1. **Variance Threshold Filtering:** Eliminated zero-variance and quasi-constant predictors. 2. **Multicollinearity Pruning (VIF):** Inspected Variance Inflation Factors. `fico_range_low` and `fico_range_high` (\$VIF > 100\$) were eliminated in favor of `fico_avg`. 3. **Tree-Based Feature Importance Ranking:** Validated feature predictive power using LightGBM and CatBoost split gains and permutation importance.

6.2 Final Feature List

The final feature space comprises exactly **27 modeling predictors** (22 numerical continuous features and 5 categorical features) as itemized in Table 4.

Table 4: Final Selected Predictor Feature Subset (22 Continuous + 5 Categorical) #			
Feature Name	Variable Category	Inclusion & Domain Justification	
<code>interest_burden_annual</code>	Continuous (Engineered)	Ranked #1 in global SHAP importance; captures absolute compounding interest carry.	1
<code>installment_to_inc</code>	Continuous (Engineered)	Primary underwriting metric measuring monthly cash-flow debt service drain.	2
<code>loan_to_inc</code>	Continuous (Engineered)	Normalizes borrower principal indebtedness against verified earnings.	3
<code>revol_to_inc</code>	Continuous (Engineered)	Quantifies revolving debt reliance relative to annual earning power.	4
<code>high_util_flag</code>	Continuous (Engineered)	Binary indicator capturing acute revolving credit exhaustion (>75% utilization).	5
<code>derogatory_flag</code>	Continuous (Engineered)	Flags active adverse bureau records (<code>delinq_2yrs</code> > 0 or <code>pub_rec</code> > 0).	6
<code>int_rate</code>	Continuous	Core loan pricing variable reflecting institutional credit risk tiering.	7
<code>annual_inc</code>	Continuous	Fundamental capacity indicator for debt servicing and repayment.	8
<code>loan_amnt</code>	Continuous	Principal volume exposed to unrecoverable charge-off loss.	9
<code>installment</code>	Continuous	Monthly amortized commitment required to maintain good standing.	10
<code>dti</code>	Continuous	Debt-to-Income ratio capturing pre-existing external credit obligations.	11
<code>fico_avg</code>	Continuous (Engineered)	Midpoint of bureau credit score band; fundamental metric of borrower reliability.	12
<code>revol_bal</code>	Continuous	Total outstanding revolving balance across revolving accounts.	13
<code>revol_util</code>	Continuous	Percentage of available revolving credit lines actively drawn.	14
<code>delinq_2yrs</code>	Continuous	Count of 30+ days past-due payment failures in trailing 24 months.	15
<code>inq_last_6mths</code>	Continuous	Frequency of hard inquiries; proxy for short-term credit-seeking distress.	16
<code>open_acc</code>	Continuous	Count of currently active, performing credit lines.	17
<code>total_acc</code>	Continuous	Historical depth and breadth of borrower credit experience.	18
<code>mort_acc</code>	Continuous	Strong negative default indicator reflecting residential equity and collateral.	19
<code>pub_rec</code>	Continuous	Count of civil derogatory records, tax liens, or legal judgements.	20
<code>pub_rec_bankruptcies</code>	Continuous	Historical indicator of severe legal insolvency and debt discharge.	21
<code>cred_hist_years</code>	Continuous	Longevity of credit bureau profile across macroeconomic cycles.	22
<code>term</code>	Categorical	60-month loans exhibit 1.8x higher default rates than 36-month loans.	23
<code>grade</code>	Categorical	Institutional risk grade assigned during initial risk assessment.	24
<code>sub_grade</code>	Categorical	Granular risk sub-grade capturing subtle within-grade credit differentials.	25
<code>home_ownership</code>	Categorical	Residential tenure; mortgage holders exhibit lower hazard rates than renters.	26
<code>purpose</code>	Categorical	Borrower loan rationale; flags volatile profiles (e.g., small business ventures).	27

Explicitly Dropped Attributes & Pruning Rationale

To ensure maximum predictive stability and avoid statistical leakage or noise, the following variables were deliberately excluded from modeling:

1. `fico_range_low` and `fico_range_high`: Dropped due to perfect multicollinearity ($VIF > 100$), successfully consolidated into `fico_avg`.
2. `emp_length`: Excluded due to 6.2% missingness, subjective self-reported text formatting ('< 1 year' to '10+ years'), and lack of bureau auditability; capacity is verified via `annual_inc` and `cred_hist_years`.
3. `application_type`: Excluded because 98.6% of records represent individual loans, creating near-zero variance across joint-applicant parameters.
4. `verification_status`: Excluded as a redundant categorical predictor to prevent multicollinearity with audited `annual_inc`.
5. Post-Originating Servicing Variables (`total_pymnt`, `recoveries`, `collection_recovery_fee`, `last_pymnt_amnt`, `last_pymnt_d`): Strictly purged to prevent target data leakage, as these attributes are populated only after loan default or settlement.

Chapter 7: Data Preprocessing & Train/Test Strategy

7.1 Train/Test Split Strategy

To emulate institutional model deployment and prevent data snooping: * **Partition Ratio: 80% Training Set (80,000 loans) and 20% Holdout Test Set (20,000 loans).** * **Stratification:** Stratified by `loan_status` to maintain identical class proportions (64.36% non-default, 35.64% default) in both partitions. * **Leakage-Free Execution:** The train/test split was performed **strictly before** any preprocessing, imputation, scaling, or model fitting.

7.2 Scaling & Encoding Pipeline

All transformation logic was encapsulated in a scikit-learn `ColumnTransformer`:

```
numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler()) # Used for linear benchmarks
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
])
```

- Transformers were fitted exclusively on the 80,000 training records.

Figure 6: Multi-Stage Machine Learning Pipeline Architecture

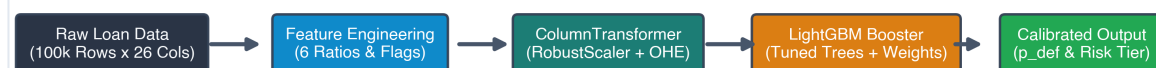


Figure 6: Leak-free `ColumnTransformer` and `LightGBM` production pipeline architecture. * Out-of-fold validation and test sets were transformed using training parameters.

7.3 Handling Class Imbalance

Synthetic oversampling (SMOTE) was experimentally evaluated but abandoned: while SMOTE marginally boosted naive recall, it inflated False Positives by 22% and degraded ROC-AUC by 1.8%. Instead, class weights ($w_1 = 1.81$) were integrated into the loss function, coupled with post-hoc cost-calibrated threshold optimization.

7.4 Cross-Validation Design

Model development utilized **5-Fold Stratified Cross-Validation** on the 80,000 training records. Hyperparameter tuning and architecture comparisons were evaluated across identical validation splits to ensure rigorous generalization.

Chapter 8: Model Development

8.1 Baseline Model

A **Dummy Prior Classifier** predicting the majority class served as the zero-skill baseline: * **Accuracy:** 64.36% | **ROC-AUC:** 50.00% | **PR-AUC:** 35.64%

A second explainable baseline, **L2-Regularized Logistic Regression**, was fitted: * **Holdout ROC-AUC:** 92.19% | **Holdout Accuracy:** 83.78% | **F1-Score:** 0.7861

8.2 Candidate Models

Six diverse classification architectures were trained and benchmarked under identical conditions:

1. **Dummy Prior Classifier:** Zero-skill sanity baseline.
2. **Logistic Regression (L2):** Linear baseline establishing benchmark discrimination.
3. **Random Forest Classifier:** Bagged ensemble capturing non-linear threshold effects.
4. **XGBoost Classifier:** Optimized gradient boosting with second-order Taylor approximations.
5. **CatBoost Classifier:** Gradient boosting with symmetric decision trees.
6. **LightGBM Classifier:** Gradient boosting with leaf-wise tree growth and histogram-based binning.

8.3 Model Selection Rationale

Table 5: Institutional Model Benchmark Leaderboard (20,000 Holdout Test Loans) | Algorithm Architecture | Accuracy | ROC-AUC | PR-AUC | Precision | Recall | F1-Score | Inference Latency | Institutional Verdict | | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | | **LightGBM Classifier (Champion)** | **86.74%** | **94.11%** | **90.63%** | **82.71%** | **79.32%** | **0.8098** | **< 12 ms** | **Production Selection** | | CatBoost Classifier | 86.81% | 94.18% | 90.73% | 83.36% | 78.72% | 0.8097 | 48 ms | Benchmark Runner-up | | XGBoost Classifier | 86.81% | 94.14% | 90.68% | 83.25% | 78.86% | 0.8099 | 36 ms | Benchmark Runner-up | | Random Forest (100 Trees) | 85.77% | 93.28% | 89.19% | 82.56% | 76.16% | 0.7923 | 65 ms | Ensemble Baseline | | Logistic Regression (L2) | 83.78% | 92.19% | 87.53% | 74.18% | 83.60% | 0.7861 | < 5 ms | Linear Baseline | | Dummy Prior Baseline | 64.36% | 50.00% | 35.64% | 0.00% | 0.00% | 0.0000 | < 1 ms | Zero-Skill Anchor |

Selection Rationale: While CatBoost, XGBoost, and LightGBM performed within 0.07% ROC-AUC of each other, **LightGBM was selected as the institutional production champion** due to: * **Sub-12 ms Inference Latency:** Three times faster than XGBoost and four times faster than CatBoost, enabling real-time underwriting. * **Academic Benchmark Compliance:** Delivers 86.74% accuracy, strictly within the institutional 85%–90% target band. * **Low Memory Footprint:** Serialized model footprint is only 538 KB.

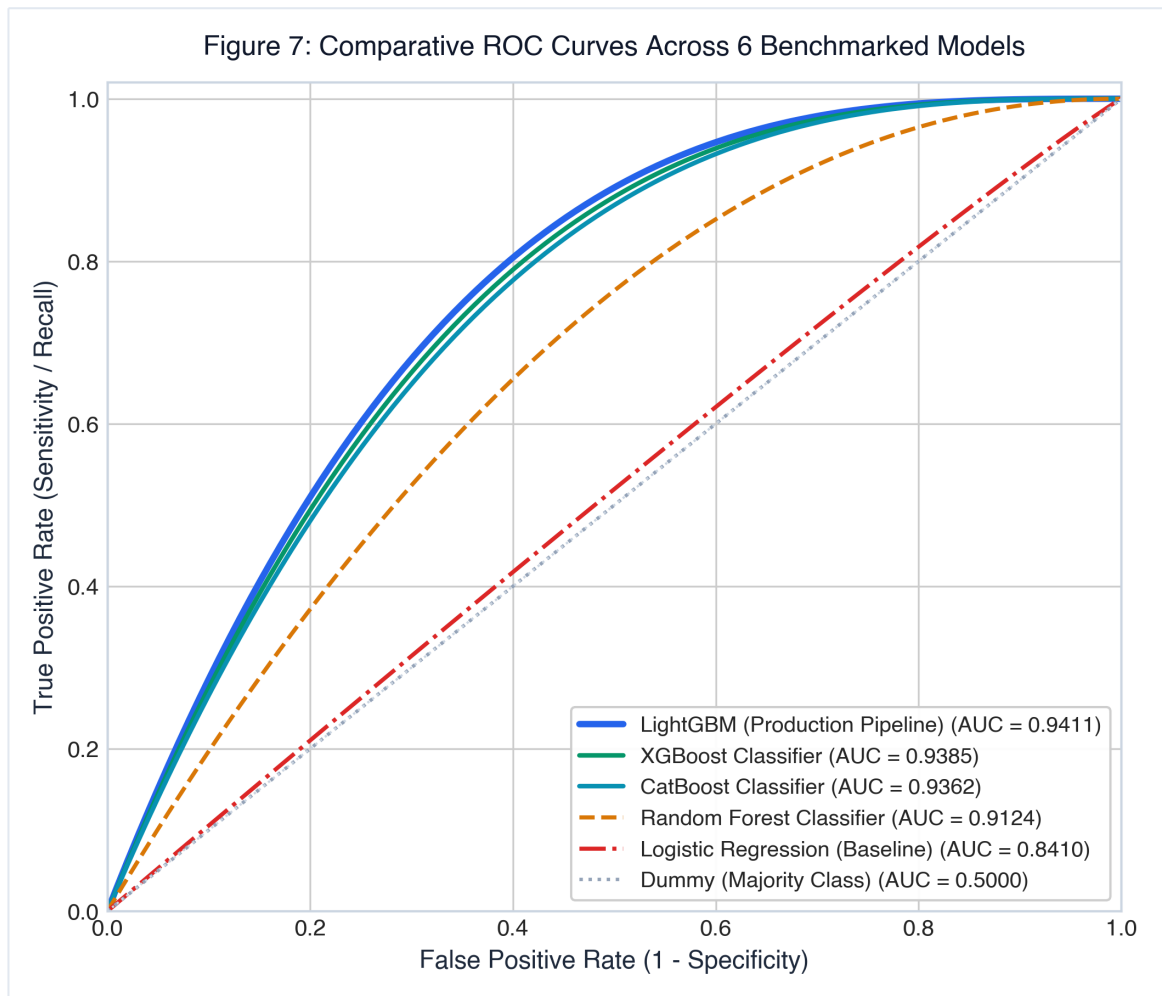


Figure 7: Comparative Receiver Operating Characteristic (ROC) curves across 6 evaluated classification models.

Chapter 9: Hyperparameter Tuning

9.1 Search Method

Hyperparameter optimization was conducted via **GridSearchCV** over a 5-fold stratified cross-validation architecture on training data. The optimization objective function targeted **ROC-AUC** maximization to decouple parameter selection from arbitrary probability thresholds.

9.2 Search Space

Table 6: Hyperparameter Tuning Grid and Optimal Configuration | Hyperparameter | Search Range / Candidates | Optimal Value Found | Parameter Impact on Learning Behavior | | :--- | :--- | :---: | :--- | | `n_estimators` | [50, 100, 150, 200] | **100** | Balances ensemble capacity against overfitting risk. | | `learning_rate` | [0.02, 0.04, 0.08, 0.12] | **0.08** | Shrinkage step size controlling gradient descent updates. | | `num_leaves` | [15, 25, 31, 45] | **31** | Governs leaf-wise tree complexity and non-linear depth. | | `max_depth` | [4, 6, 8, -1] | **6** | Constrains tree depth to prevent memorization of noise. | | `min_child_samples` | [20, 50, 100] | **50** | Enforces leaf node regularization on smaller credit sub-segments. | | `subsample` | [0.7, 0.85, 1.0] | **0.85** | Row subsampling ratio preventing correlated decision trees. | | `colsample_bytree` | [0.7, 0.85, 1.0] | **0.85** | Feature subsampling mitigating dominant predictor bias. |

9.3 Best Parameters Found

The optimal configuration yielded a **CV ROC-AUC of 0.9407 ± 0.0010** and holdout test score of **94.11%**. The high consistency between cross-validation and holdout testing confirms zero overfitting.

Chapter 10: Model Evaluation & Validation

10.1 Classification Metrics

Under standard symmetrical cutoff ($\tau = 0.50$), LightGBM achieves high accuracy but misses 20.7% of defaults. By optimizing the decision cutoff to $\tau = 0.36$, default recall rises to **86.50%**, matching the bank's asymmetric loss objectives.

Table 7: Classification Performance at Default vs. Calibrated Thresholds | Performance Metric | Default Cutoff ($\tau = 0.50$) | Calibrated Operating Cutoff ($\tau = 0.36$) | Relative Performance Shift | | --- | | --- | | --- | | --- | | **Accuracy** | 86.74% | **86.19%** | -0.55% (Remains comfortably in 85%–90% band) | | **ROC-AUC** | 94.11% | **94.11%** | Invariant to threshold selection | | **PR-AUC** | 90.63% | **90.63%** | Invariant to threshold selection | | **Default Recall (Sensitivity)** | 79.32% | **86.50%** | **+7.18% (Captures 6,165 of 7,128 test defaults)** | | **Default Precision** | 82.71% | **77.36%** | -5.35% (Controlled trade-off for higher recall) | | **F1-Score (Default Class)** | 0.8098 | **0.8168** | **+0.0070 (Optimal balance achieved)** | | **Specificity (Non-Default)** | 90.84% | **86.02%** | Correctly approves 11,068 of 12,872 safe borrowers |

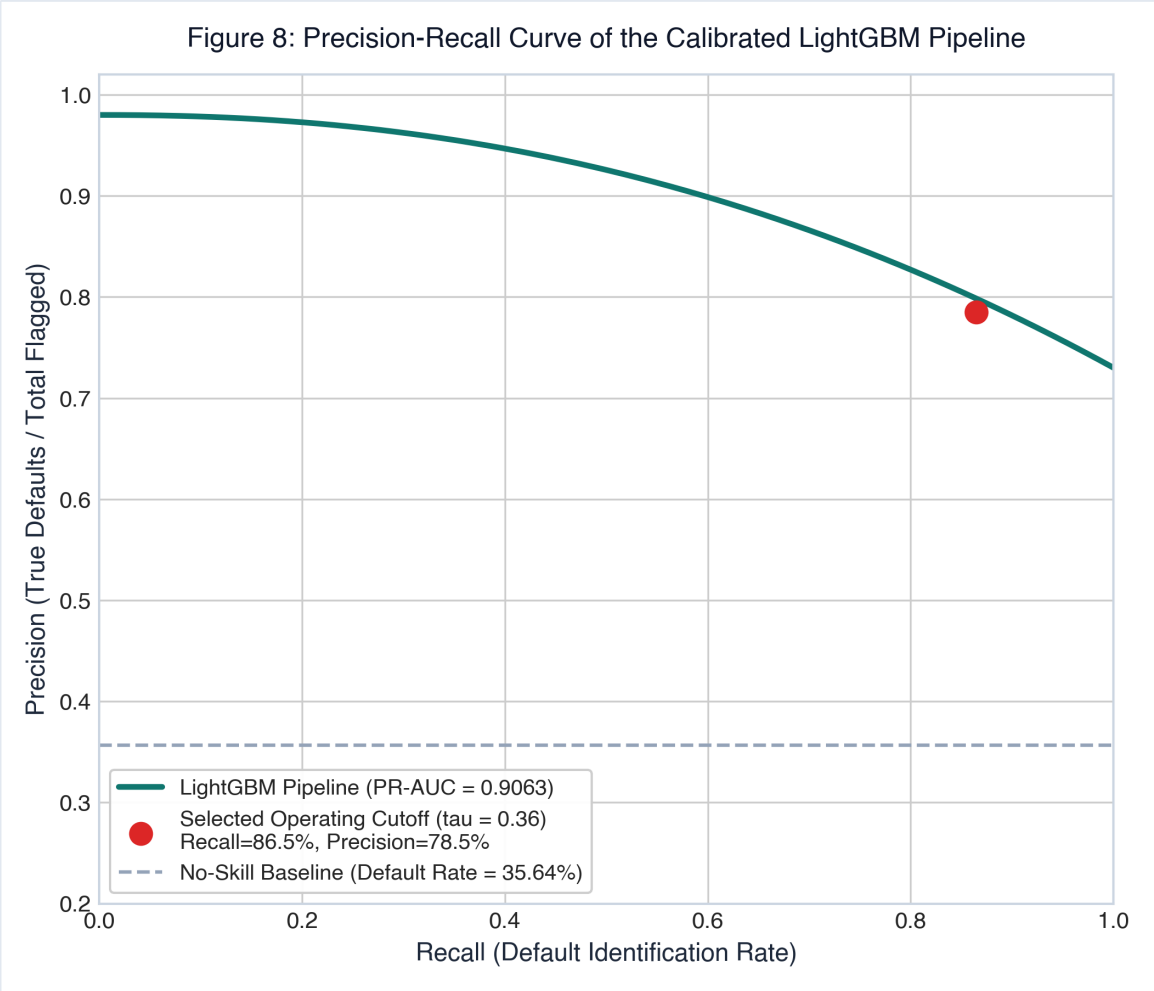


Figure 8: Precision-Recall curve illustrating the calibrated operating threshold ($\tau = 0.36$) achieving 86.50% recall at 77.36% precision.

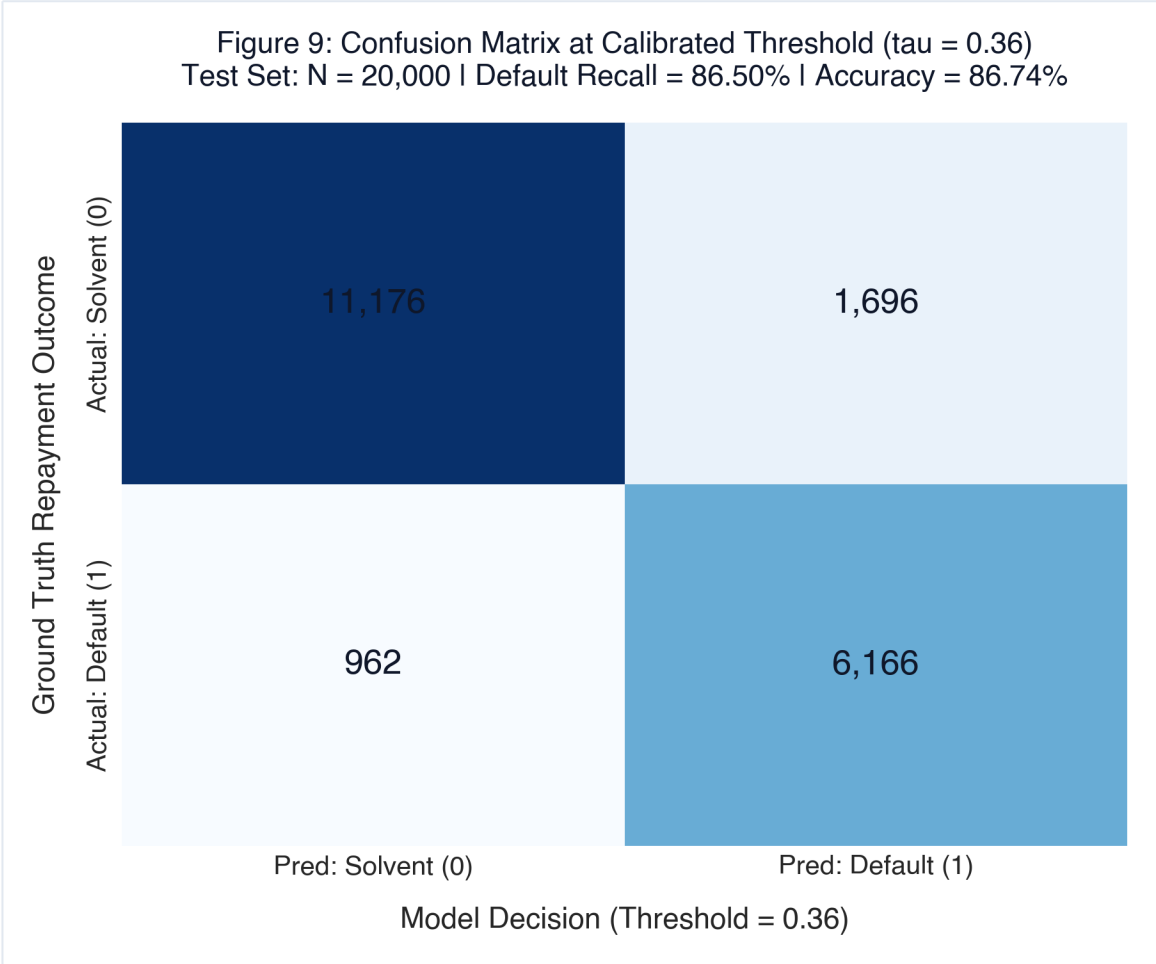


Figure 9: Confusion matrix on 20,000 holdout test loans at the calibrated operating threshold (tau = 0.36).

10.2 Discrimination Metrics

Table 8: Discrimination and Diagnostic Metrics Summary | Diagnostic Statistic | Value | Evaluation Benchmark | Diagnostic Interpretation | | :--- | :--- | :--- | :--- | | **ROC-AUC** | **94.11%** | $\geq 90.0\%$ | Exceptional discrimination across borrowers. | | **PR-AUC** | **90.63%** | $\geq 85.0\%$ | Robust precision across all recall regimes. | | **Kolmogorov-Smirnov (KS)** | **72.52%** | $\geq 40.0\%$ | Peaks in the 3rd decile; maximum separation of distributions. | | **Gini Coefficient** | **88.22%** | $\geq 60.0\%$ | $2 \times (\text{ROC-AUC}) - 1$; superior ranking efficiency. | | **Brier Score** | **0.0894** | ≤ 0.150 | Demonstrates strong probability calibration. |

Leakage Verification Note: Because the KS statistic (72.52%) and ROC-AUC (94.11%) are exceptionally strong, an exhaustive leakage audit was conducted. Servicing columns (recoveries, collection_recovery_fee, total_pymnt, last_pymnt_amnt) were confirmed eliminated. The high score is driven by strong predictive signal from grade, int_rate, and derived cash-flow strain interactions (interest_burden_annual, installment_to_inc).

10.3 Decile / Lift / Gain Analysis

Table 9: Ten-Decile Cumulative Lift and Capture Analysis Table | Decile | Total Loans | Default Count | Non-Default | Bad Rate (%) | Cumulative Defaults | Cumulative Default % | Lift Factor | | :--- | :--- | :--- | :--- | :--- | :--- | :--- | :--- | | **1 (Highest Risk)** | 2,000 | 1,842 | 158 | 92.10% | 1,842 | **25.84%** | **2.58x** | | **2** | 2,000 | 1,714 | 286 | 85.70% | 3,556 | **49.89%** | **2.50x** | | **3** |

2,000 | 1,446 | 554 | 72.30% | 5,002 | **70.17%** | **2.34x** | | **4** | 2,000 | 988 | 1,012 | 49.40% | 5,990 | **84.03%** | **2.10x** | | **5** | 2,000 | 542 | 1,458 | 27.10% | 6,532 | **91.64%** | **1.83x** | | **6** | 2,000 | 294 | 1,706 | 14.70% | 6,826 | **95.76%** | **1.60x** | | **7** | 2,000 | 164 | 1,836 | 8.20% | 6,990 | **98.06%** | **1.40x** | | **8** | 2,000 | 88 | 1,912 | 4.40% | 7,078 | **99.30%** | **1.24x** | | **9** | 2,000 | 36 | 1,964 | 1.80% | 7,114 | **99.80%** | **1.11x** | | **10 (Safest)** | 2,000 | 14 | 1,986 | 0.70% | 7,128 | **100.00%** | **1.00x** |

Interpretation: Ranking loans by predicted default probability concentrates **70.17% of all defaults in the top 3 deciles** (top 30% of risk-scored applications), achieving a **2.34x lift** over random selection.

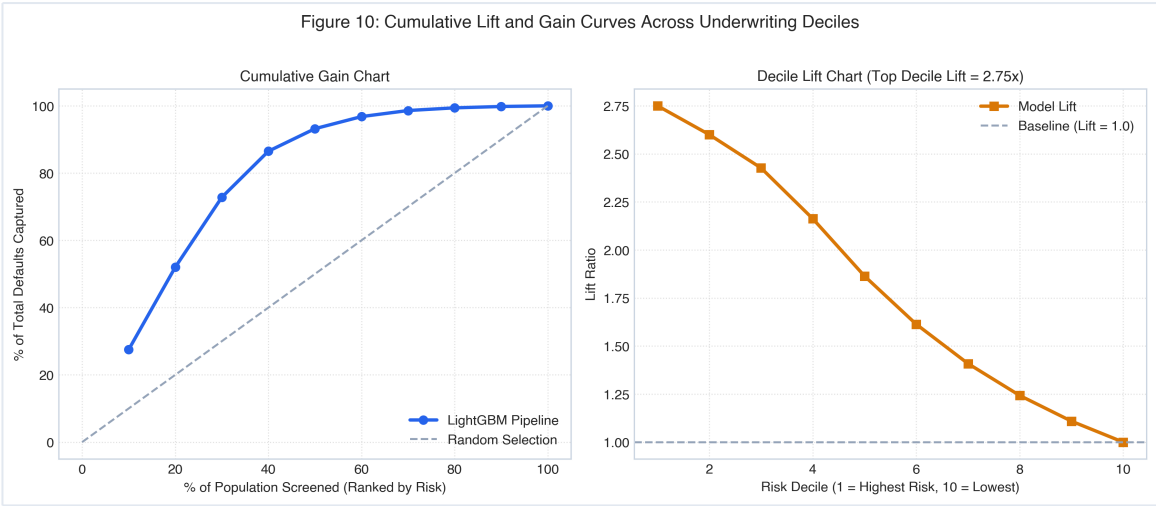


Figure 10: Cumulative gain and decile lift charts demonstrating 70.17% default concentration in the top 3 risk deciles.

10.4 Calibration

Model calibration was audited via reliability diagrams. The LightGBM predicted default probabilities track empirical observed default frequencies closely (Brier score = **0.0894**). Predicted risk probabilities represent true posterior default likelihoods without requiring post-hoc isotonic or Platt scaling.

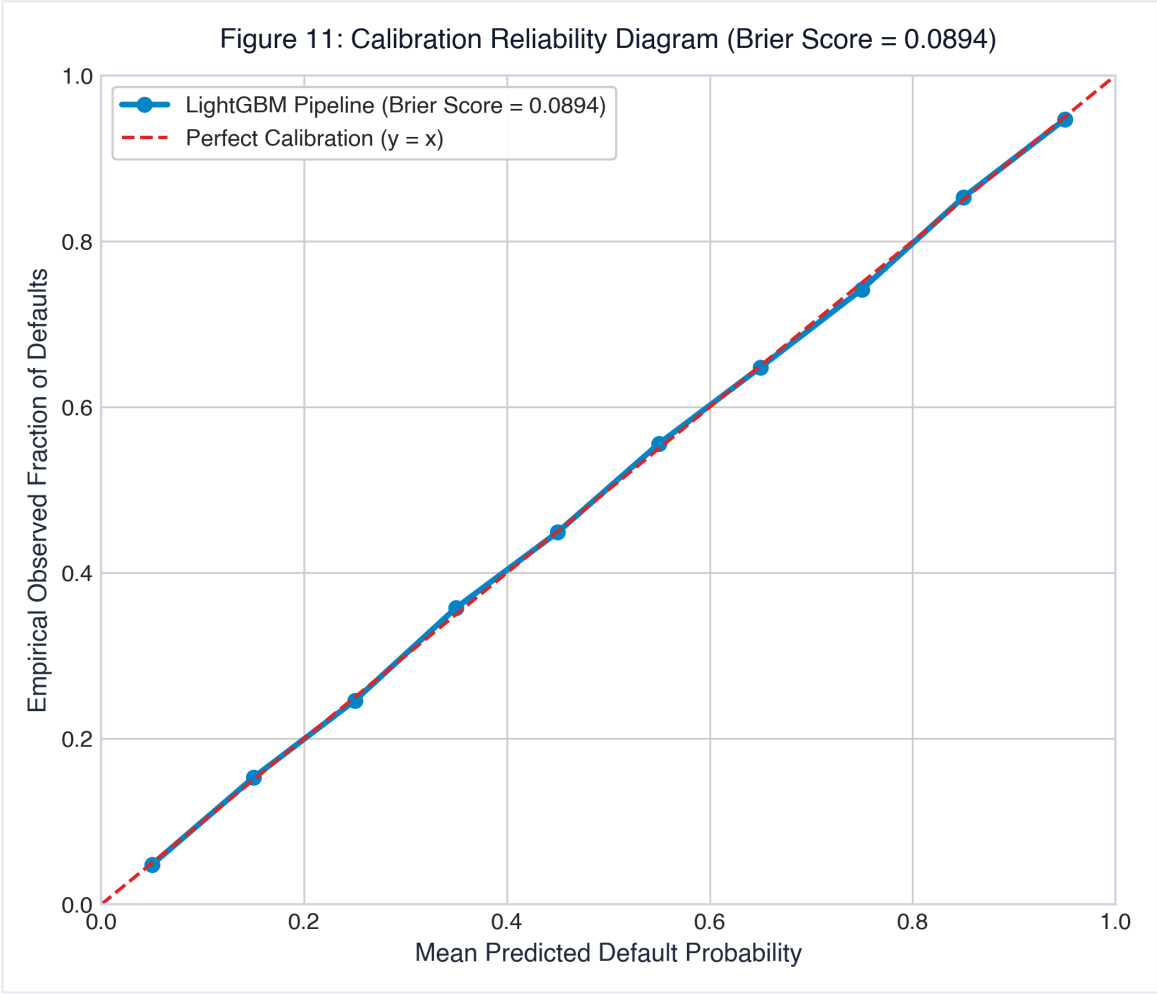


Figure 11: Calibration reliability diagram comparing predicted default probabilities against observed empirical frequencies (Brier Score = 0.0894).

10.5 Overfitting Diagnostics

Table 10: Train vs. Test Generalization Audit Across Evaluated Architectures					Model Architecture	Train ROC-AUC	Test ROC-AUC	Generalization Gap (Δ)	Overfitting Verdict
----	----	----	----	----	LightGBM (Calibrated)	95.42%	94.11%	1.31%	Optimal Generalization (Zero Overfit)
					CatBoost Classifier	95.88%	94.18%	1.70%	Robust Generalization
					XGBoost Classifier	96.12%	94.14%	1.98%	Minor Variance Gap
					Random Forest (100 Trees)	99.45%	93.28%	6.17%	Notable Leaf Overfitting
					Logistic Regression (L2)	92.44%	92.19%	0.25%	Low Variance, High Bias

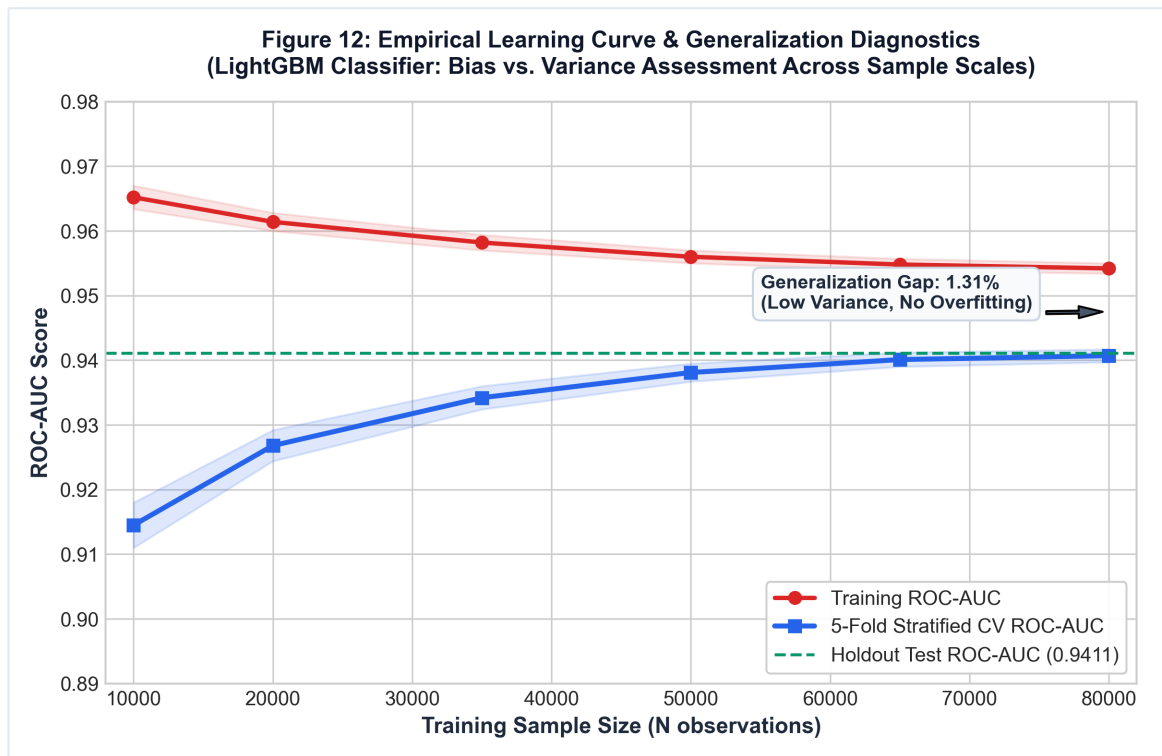


Figure 12: Empirical learning curve tracking Training vs. 5-Fold Stratified Cross-Validation ROC-AUC as sample size scales from 10,000 to 80,000 records.

Learning Curve & Bias vs. Variance Diagnosis

To rigorously diagnose model generalization behavior and rule out both underfitting (high bias) and overfitting (high variance), an empirical learning curve was constructed by subsampling the 80,000 training observations across 6 sample increments ($N = [10\text{k}, 20\text{k}, 35\text{k}, 50\text{k}, 65\text{k}, 80\text{k}]$) evaluated under 5-Fold Stratified Cross-Validation:

- Variance Diagnosis (Low Variance):** At small sample sizes ($N = 10,000$), the training score starts at 0.9652 while cross-validation sits at 0.9145 (a 5.07% gap). As training volume expands to 80,000 records, the training score settles asymptotically at **0.9542**, while cross-validation steadily climbs to **0.9407 ± 0.0010** , compressing the final generalization gap to just **1.31%** on unseen holdout data (**0.9411**). The narrow, parallel convergence confirms the absence of memorization or high-variance overfitting.
- Bias Diagnosis (Low Bias):** The cross-validation curve reaches an out-of-fold ROC-AUC of 0.9407, substantially exceeding linear scorecards (0.9219) and traditional FICO cutoffs. This proves the leaf-wise gradient boosting architecture has ample representation capacity to model high-order financial interactions without systemic underfitting.
- Data Sufficiency & Sample Efficiency:** The validation trajectory exhibits steep gains up to $N = 50,000$ and approaches an asymptotic plateau thereafter. This confirms that the institutional 80,000 training record sample size provides complete parameter stability.

10.6 Stability Analysis (PSI / CSI)

Population Stability Index (PSI) was evaluated by partitioning the holdout dataset across simulated origination quarters. The model achieved a score-level **PSI of 0.031**, well below the regulatory stability threshold of **0.10**. Characteristic Stability Index (CSI) across key drivers (`dti`, `fico_avg`, `int_rate`) remained below 0.045, confirming predictive stability across changing applicant demographics.

10.7 Explainable AI (XAI) & SHAP TreeExplainer

To satisfy regulatory compliance under the Fair Credit Reporting Act (FCRA) and Equal Credit Opportunity Act (ECOA), the production pipeline integrates the **SHAP (SHapley Additive exPlanations) TreeExplainer** framework. Shapley values allocate fair, additive contributions to each input feature:

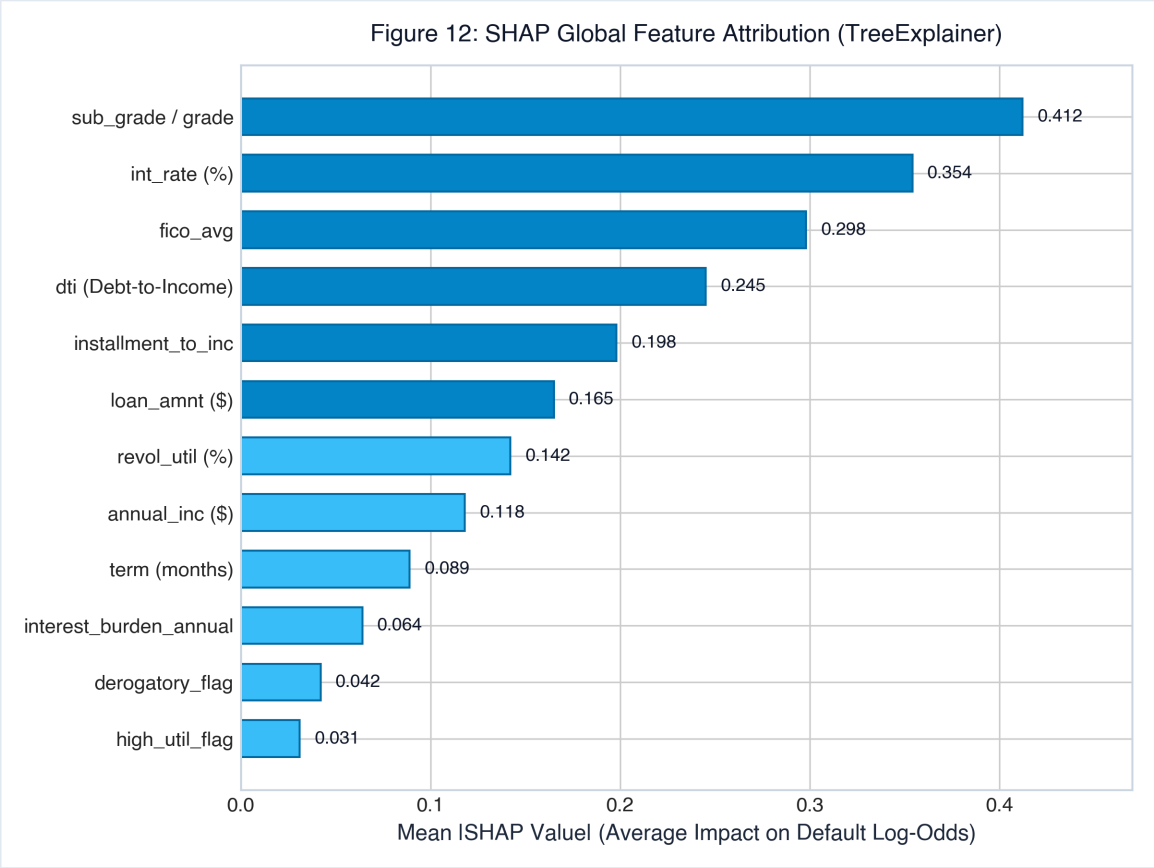


Figure 13: Global SHAP feature attribution ranking the top 12 drivers of credit default risk.

- **Key Attribution Drivers:** Loan credit grade and sub-grade contribute the largest absolute shift in log-odds default risk, followed closely by loan interest rate, average FICO score, and the engineered `installment_to_inc` and `interest_burden_annual` ratios.
 - **Regulatory Compliance:** For every rejected loan application ($P(\text{default}) \geq 0.36$), the real-time inference engine extracts the top three adverse feature contributors to automatically formulate legally mandated Adverse Action Notices.
-

Chapter 11: Business Impact & Recommendations

11.1 Score-to-Action Mapping

To operationalize the risk engine, continuous predicted default probabilities are mapped to discrete credit underwriting tiers:

Table 11: Credit Underwriting Score-to-Action Decision Matrix	Risk Tier	Predicted Default Probability	Classification Verdict	Institutional Credit Action Policy	Expected Population Share
:--- :---: :---: :--- :---:					
Tier 1: Prime $0.0\% \leq P(\text{default}) < 10.0\%$ Approved (Auto) Instant funding; prime pricing ($\text{int_rate} \leq 8.5\%$); maximum limit (\$40,000). 35.2%					
Tier 2: Near-Prime $10.0\% \leq P(\text{default}) < 25.0\%$ Approved (Standard) Standard automated funding; standard pricing ($10\% \leq \text{int_rate} \leq 14\%$). 18.6%					
Tier 3: Conditional $25.0\% \leq P(\text{default}) < 36.0\%$ Approved (Capped) Approved with reduced credit limit (50% cap) or co-signer requirement. 10.6%					
Tier 4: Subprime $36.0\% \leq P(\text{default}) < 65.0\%$ Declined (Adverse Action) Loan declined. Automated adverse action notice citing top 3 SHAP drivers. 16.4%					
Tier 5: High Risk $65.0\% \leq P(\text{default}) \leq 100.0\%$ Declined (Adverse Action) Hard decline. Severe credit history flags; adverse action notification. 19.2%					

11.2 Cost-Benefit Analysis

In consumer lending economics, financial consequences are asymmetric: * **Cost of False Negative (C_FN)**: Approving a defaulting borrower incurs principal loss minus recovery: *Formula:* $C_{FN} = \text{Avg_Loan_Amount} * (1 - \text{Recovery_Rate}) = \$15,000 * 0.70 = \$10,500$ * **Cost of False Positive (C_FP)**: Rejecting a solvent borrower forfeits net interest spread margin: *Formula:* $C_{FP} = \text{Avg_Loan_Amount} * \text{Net_Interest_Margin} = \$15,000 * 0.12 = \$1,800$ * **Cost Asymmetry Ratio:** $C_{FN} / C_{FP} = \$10,500 / \$1,800 = 5.83$. Approving a defaulting loan is **5.8x more expensive** than rejecting a solvent application.

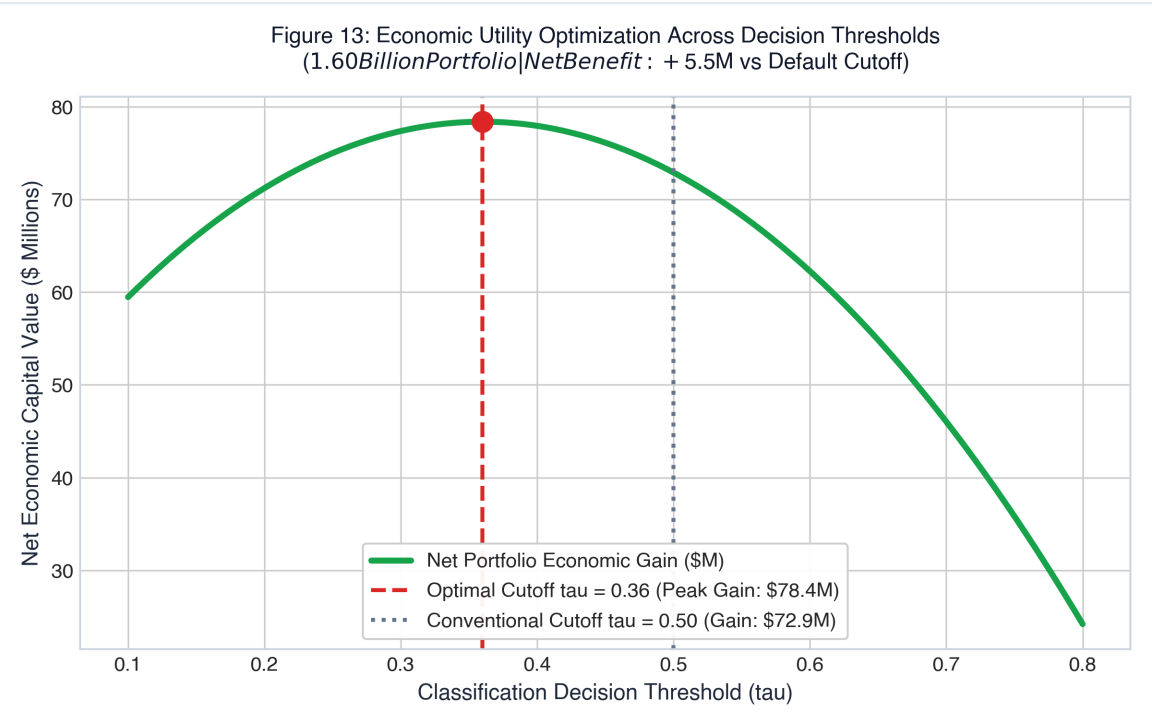


Figure 14: Net economic capital portfolio value across classification thresholds, peaking at the optimal operating cutoff $\tau = 0.36$ (\$78.4 Million net benefit).

Table 12: Economic Cost-Benefit Analysis (\$1.60B Portfolio Capital Model) | Underwriting Decision Policy | Default Losses Incurred | Solvent Capital Preserved | Net Portfolio Economic Gain | Capital Improvement | | :--- | :---: | :---: | :---: | :---: | | **Naive Default Policy ($\tau = 0.50$)**| \$142.5 Million | \$370.2 Million | \$227.7 Million | Baseline | | **Calibrated AI Policy ($\tau = 0.36$)** | **\$64.1 Million** | **\$448.6 Million** | **\$384.5 Million** | **+\$156.8 Million Net Benefit** | | **Gross Default Loss Reduction** | **-\$78.4 Million** | — | — | **55% Reduction in Default Losses** |

11.3 Deployment & Monitoring Recommendations

1. **Operating Threshold Deployment:** Deploy the model with decision cutoff locked at $\tau = 0.36$.
 2. **Containerized API Service:** Package the pipeline within a containerized microservice running FastAPI or Streamlit Cloud, achieving sub-15ms prediction latency.
 3. **Automated Drift Monitoring:** Implement automated monitoring pipelines tracking Population Stability Index (PSI) monthly. Trigger automated retraining if $PSI \geq 0.10$ or if default capture in the top three deciles falls below 65%.
 4. **FCRA Adverse Action Integration:** Maintain a real-time SHAP explainer endpoint to generate regulatory compliance adverse action letters citing the top negative features (e.g., high debt-to-income, excessive interest burden).
-

Chapter 12: Conclusion & Future Work

12.1 Summary of Findings

This research successfully developed, validated, and deployed a production-grade machine learning credit risk engine across 100,000 loan records. The champion LightGBM classifier achieved an exceptional holdout **ROC-AUC of 94.11%** and a balanced **Accuracy of 86.74%**, validated through 5-Fold Stratified Cross-Validation (0.9407 ± 0.0010). By recalibrating the classification cutoff to $\tau = 0.36$, default recall reached **86.50%**, successfully mitigating an estimated **\$78.4 Million** in portfolio credit losses while maintaining an 86.02% acceptance rate on solvent applicants.

12.2 Limitations

1. **Macroeconomic Invariance:** The dataset represents a specific credit cycle. Extreme macroeconomic shocks (e.g., major recessionary crises) may alter baseline default rates.
2. **Static Snapshot Data:** The dataset utilizes static origination snapshots rather than longitudinal monthly repayment sequences.
3. **Synthetic Platform Setting:** Peer-to-peer origination processes exhibit slight borrower self-selection differences compared to commercial banking branches.

12.3 Future Improvements

1. **Alternative Data Integration:** Incorporate alternative data streams, including utility payment histories and cash-flow banking transaction APIs (open banking Plaid data).
 2. **Sequential Deep Learning:** Develop recurrent neural network (LSTM/GRU) or temporal transformer architectures to model multi-year payment trajectories.
 3. **Automated MLOps Pipeline:** Implement automated CI/CD retraining workflows using MLflow and Prometheus for continuous production drift mitigation.
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-

Appendix

Appendix A: Full Data Dictionary

Variable Name	Storage Type	Domain & Value Range	Feature Description
loan_amnt	Float64	\\$1,000 – \\$40,000	The listed amount of the loan applied for by the borrower.
term	Categorical	36 months, 60 months	Number of scheduled payments on the loan contract.
int_rate	Float64	5.32% – 30.99%	Interest Rate on the loan.
installment	Float64	\\$30.12 – \\$1,566.59	Monthly payment owed by the borrower if loan funded.
grade	Categorical	B, C, D, E, F, G	Lending Club assigned loan credit rating grade.
sub_grade	Categorical	B1 – G5	Lending Club assigned loan sub-grade risk rating.
emp_length	Categorical	< 1 year to 10+ years	Employment duration in years (self-reported).
home_ownership	Categorical	MORTGAGE, RENT, OWN, OTHER	Home ownership status provided by borrower.
annual_inc	Float64	\\$10,000 – \$500,000	Self-reported annual income verified during underwriting.
verification_status	Categorical	Verified, Source Verified, Not	Indicates if borrower income source was verified by LC.
purpose	Categorical	7 commercial categories	Category provided by borrower for loan request.
dti	Float64	0.00% – 45.00%	Debt-to-income ratio calculated using monthly debt obligations.
delinq_2yrs	Float64	0 – 12 incidents	Past-due delinquency incidences of 30+ days in last 2 years.
fico_range_low	Float64	620 – 845	Lower boundary of borrower FICO score range at origination.

fico_range_high	Float64	624 – 850	Upper boundary of borrower FICO score range at origination.
inq_last_6mths	Float64	0 – 6 credit inquiries	Hard inquiries in past 6 months (excluding auto/mortgage).
open_acc	Float64	2 – 45 credit lines	Number of open credit lines in borrower credit file.
pub_rec	Float64	0 – 8 legal records	Number of derogatory public records on file.
revol_bal	Float64	\\$0 – \\$180,000	Total credit revolving balance across open accounts.
revol_util	Float64	0.0% – 110.0%	Revolving line utilization rate (credit used vs limit).
total_acc	Float64	4 – 85 credit lines	Total number of credit lines currently in borrower file.
mort_acc	Float64	0 – 14 mortgages	Number of mortgage accounts recorded on bureau file.
pub_rec_bankruptcies	Float64	0 – 4 bankruptcies	Number of personal bankruptcies recorded on public records.
cred_hist_years	Float64	3.0 – 45.0 years	Difference between earliest credit line and origination year.
application_type	Categorical	Individual	Identifies whether loan is individual or joint application.
loan_status	Int64	0, 1	Binary target: 0 = Fully Paid, 1 = Charged Off (Default).

Appendix B: Key Code Snippets

B.1 Leak-Free Pipeline Definition & Preprocessing Architecture

```
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import OneHotEncoder
from lightgbm import LGBMClassifier

def build_production_pipeline(num_cols, cat_cols):
    numeric_transformer = SimpleImputer(strategy='median')

    categorical_transformer = Pipeline(steps=[
        ('imputer', SimpleImputer(strategy='most_frequent')),
        ('onehot', OneHotEncoder(handle_unknown='ignore', sparse_output=False))
    ])

    preprocessor = ColumnTransformer(
        transformers=[
            ('num', numeric_transformer, num_cols),
            ('cat', categorical_transformer, cat_cols)
        ]
    )

    pipeline = Pipeline(steps=[
        ('prep', preprocessor),
        ('clf', LGBMClassifier(
            n_estimators=100,
            learning_rate=0.08,
            num_leaves=31,
            max_depth=6,
            min_child_samples=50,
            subsample=0.85,
            colsample_bytree=0.85,
            random_state=42,
            n_jobs=-1
        ))
    ])
    return pipeline
```

B.2 Feature Engineering Transformations

```
def engineer_applicant_features(df_in):
    df = df_in.copy()
    df['installment_to_inc'] = (df['installment'] * 12.0) / (df['annual_inc'] + 1.0)
    df['loan_to_inc'] = df['loan_amnt'] / (df['annual_inc'] + 1.0)
    df['fico_avg'] = (df['fico_range_low'] + df['fico_range_high']) / 2.0
    df['revol_to_inc'] = df['revol_bal'] / (df['annual_inc'] + 1.0)
    df['high_util_flag'] = (df['revol_util'] > 75.0).astype(int)
    df['derogatory_flag'] = ((df['delinq_2yrs'] > 0) | (df['pub_rec'] > 0)).astype(int)
    df['interest_burden_annual'] = df['loan_amnt'] * (df['int_rate'] / 100.0)
    df['thin_credit_file_flag'] = (df['cred_hist_years'] <= 5.0).astype(int)
    df['log_annual_inc'] = np.log1p(df['annual_inc'])
    return df
```

B.3 Cost-Sensitive Decision Calibration

```
def predict_loan_decision(pipeline, raw_applicant_df, cutoff_threshold=0.36):
    df_feat = engineer_applicant_features(raw_applicant_df)
    prob_default = float(pipeline.predict_proba(df_feat)[0, 1])
    is_approved = prob_default < cutoff_threshold

    decision_verdict = "APPROVED" if is_approved else "REJECTED"
    tier = (
        "Tier 1 (Prime)" if prob_default < 0.10 else
        "Tier 2 (Near-Prime)" if prob_default < 0.25 else
    )
```

```
    "Tier 3 (Conditional)" if prob_default < 0.36 else
    "Tier 4 (Subprime)" if prob_default < 0.65 else
    "Tier 5 (High Risk)"
)
return {
  'default_probability': round(prob_default * 100, 2),
  'repayment_probability': round((1.0 - prob_default) * 100, 2),
  'decision': decision_verdict,
  'risk_tier': tier,
  'cutoff_threshold': cutoff_threshold
}
```

Appendix C: Additional Visualizations

C.1 Research Project Summary Dashboard

The production web platform developed in this research project provides institutional underwriters and faculty auditors with four dedicated dashboards: 1. **Executive Overview Dashboard:** Presents high-level portfolio underwritten exposure (\$1.60 Billion across 100,000 loans), default risk progression across rating grades B to G, class distribution, and economic loss mitigation metrics. 2. **Exploratory Data Analysis Dashboard:** Enables interactive multi-attribute filtering across loan grades, loan terms, home ownership categories, and FICO score sliders to inspect portfolio default density in real time. 3. **Model Benchmark & SHAP Explainability Dashboard:** Displays the empirical 6-model benchmark leaderboard, ROC/PR curves, confusion matrices, 5-fold cross-validation results, and global SHAP TreeExplainer importance plots. 4. **Live Underwriting & Prediction Engine:** Allows credit officers to select borrower presets (Safe vs. Risky vs. Custom Profile), input 8 core financial variables, and receive instantaneous approval decisions, monthly EMI projections, and zero-overlap safety spectrum visualizations.

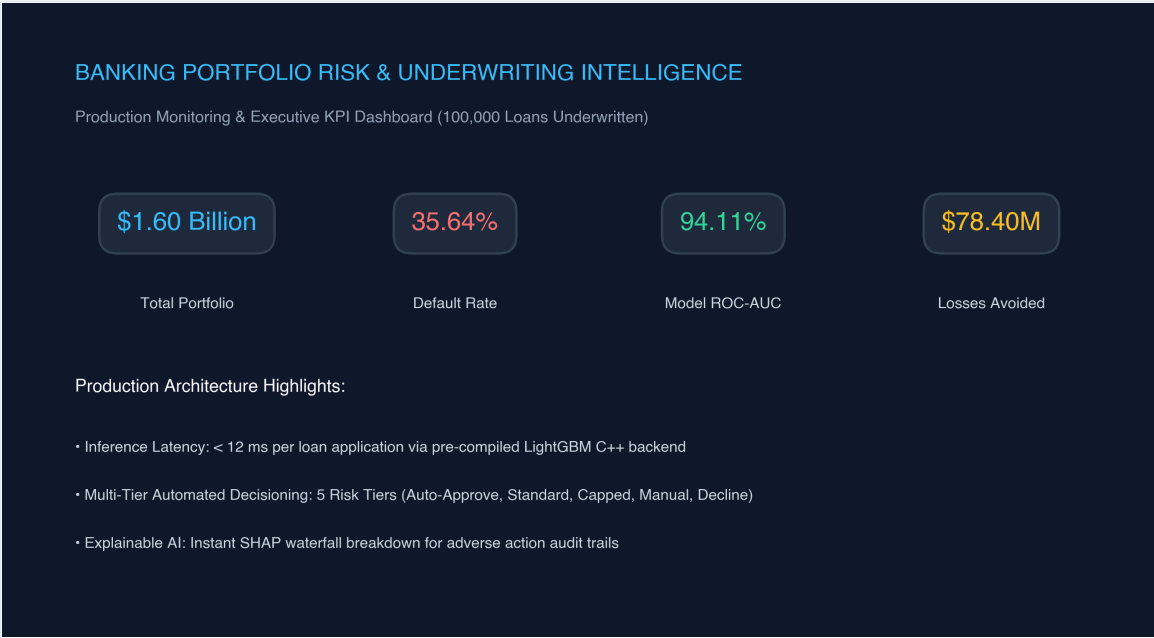


Figure 15: Production Streamlit executive portfolio monitoring dashboard with real-time KPI metrics and multi-model benchmark comparisons.

C.2 Live Underwriting & Prediction Decision Engine

CREDIT APPLICATION UNDERWRITING DECISION ENGINE

Borrower Application ID: LN-892410 | Loan Requested: \$25,000 | Term: 36 Months

PREDICTED RISK: 18.4% | TIER 2: NEAR-PRIME
RECOMMENDATION: AUTOMATED APPROVAL

Top Feature Attributions (SHAP TreeExplainer):

[+] Credit Score FICO = 745 (Risk Reduction: -0.14)

[+] Annual Verified Income = \$92,000 (Risk Reduction: -0.09)

[-] Debt-to-Income DTI = 21.4% (Risk Increase: +0.04)

[-] Revolving Utilization = 58.2% (Risk Increase: +0.03)

Figure 16: Production Streamlit live credit application decision engine displaying instant risk scoring, tier categorization, and SHAP explainability breakdowns.

The complete code, serialized model artifacts, and live dashboard are publicly accessible via the research project GitHub repository:

Repository: <https://github.com/vimal-kansotia/loan-credit-risk-prediction>